Highlights

Provenance-Controlled Graph Augmentation for Auditable Entity–Relation Extraction

- Provenance controls restrict graph hints to eligible training-derived context.
- Evidence gates separate candidate generation from source-linked output acceptance.
- Relation-F1 gains recur on ADE and CoNLL04 across runs and overlap exclusion.

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ABSTRACT

Drug–adverse-effect extraction from medical text requires traceable relations. Graph-augmented extraction can retrieve useful associations, but a retrieved association is not necessarily a fact stated in the current sentence, and fluent generators make it difficult to trace a predicted relation to its source span and audit why it was accepted. We develop a provenance-preserving, evidence-gated framework that separates graph-informed candidate generation from output acceptance. A graph derived only from training annotations supplies bounded concept and relation hints to QLoRA-adapted Qwen3-4B extractors; deterministic span reconstruction, entity acceptance, and relation verification then retain only source-linked, type-compatible assertions. ADE serves as the primary medical benchmark, with CoNLL04 providing a supplementary general-domain evaluation under separate training and a shared strict character-span protocol. The completed tests show that the provenance-gated system improves relation F1 over its matched source-only generator on both benchmarks and records explicit acceptance decisions. The relation improvement persists in two additional training repetitions on both datasets and after exact CoNLL04 train–test text overlaps are excluded. These results demonstrate that controlled graph augmentation can improve relation extraction over matched source-only generation while maintaining explicit source linkage and inspectable acceptance decisions, thereby providing an auditable foundation for human-in-the-loop knowledge construction in safety-sensitive applications.

1. Introduction

Structured information extraction connects unstructured text to searchable entities, typed relations, and downstream knowledge systems. In applications related to safety, an extracted association must also be inspectable: users need to know which mention was identified, which sentence supported the relation, and which transformations produced the final assertion. Drug-related adverse-effect extraction from medical text is the primary application in this study. A model that recognizes a drug and an adverse effect but invents an association between them can create a misleading record even when its output is syntactically valid. General-domain extraction provides a complementary test of the same technical issue across a broader set of entity and relation labels.

Neural entity and relation models provide effective approaches to the recognition problem [1, 2]. Generative models offer a flexible alternative in which a common structured output format can be reused across label inventories. However, fluent generation does not guarantee that the predicted objects are supported by their input [3]. This distinction matters when extracted triples are inserted into a graph: errors become available to later retrieval or reasoning stages, potentially propagating beyond the sentence that generated them.

Knowledge graphs can contribute typed domain priors [4], and retrieval augmentation can expose those priors without requiring all knowledge to be stored in model parameters [5]. Their use introduces a specific authority problem. A relation observed in a training sentence may suggest what to search for, but it does not establish that the same relation holds in a new sentence. A training graph may also return an example's own annotation or reveal content shared across data splits. The useful question is therefore not only whether graph context increases recall, but whether its eligibility and influence can be controlled and audited.

We address this problem with a staged architecture. Separate source-only, exact-anchor, and high-recall graph-conditioned generators establish a matched comparison of input conditions. A deterministic relation verifier checks structural and source-evidence requirements. A complementary entity gate uses agreement, source-matched anchors, or verified relation endpoints to accept a subset of the graph-conditioned candidates. The final system combines the accepted entities with verified relations whose endpoints survive. The architecture uses existing representation and adaptation methods, but introduces a provenance-constrained augmentation–acceptance protocol: graph records may influence candidate generation, yet candidates must pass explicit source-linkage and type checks before entering the output. This separates the authority of retrieved context from acceptance semantics and makes both decisions independently inspectable.

ADE [6] is the primary medical benchmark, addressing drug–adverse-effect extraction with two entity labels and one relation label. CoNLL04 [7] provides a supplementary general-domain evaluation with four entity labels and five directed relation labels, testing applicability beyond the medical task under different entity roles and relation structures. Both use English sentence-level inputs, making it possible to examine the extraction and verification mechanisms without importing a long-document windowing problem. They are trained and evaluated separately; this is not a cross-dataset zero-shot transfer experiment.

The contributions are threefold:

- A provenance-constrained augmentation policy that separates graph eligibility from candidate generation, uses training annotations only, excludes current-example-only support, and clears graph hints for exact split overlaps.

- An evidence-gated acceptance mechanism that decouples generation from output acceptance, combining source-span reconstruction, entity selection, relation verification, and endpoint consistency with inspectable decisions.
- Empirical evidence of relation-extraction gains over matched source-only generation on the primary medical and supplementary general-domain benchmarks, retained across two additional training runs and exact-overlap exclusion, together with ablations that characterize how graph context and acceptance gates affect precision and recall.

2. Related work

2.1. Joint entity and relation extraction

Entity recognition must resolve both mention boundaries and types; relation extraction additionally depends on recovering the correct directed endpoints [1]. Global consistency constraints have a long history in this setting, including the formulation associated with the CoNLL04 benchmark [7]. SpERT enumerates spans and jointly models entities and relations using a pretrained transformer [2]. It is a particularly relevant comparison because it performs the same joint task without requiring a generative JSON interface. Our experiments retrain SpERT from a clean base encoder on the same training partitions rather than using a released task checkpoint that may already include development data.

Recent joint extractors explore complementary output representations. OneRel casts extraction as fine-grained triple classification, UniRel unifies entity and relation interactions, and set-prediction networks treat the output as an unordered collection of relational tuples [8–10]. Boundary-oriented methods either assemble mention boundaries or learn them through regression, whereas position-aware models emphasize endpoint structure and relation representations [11–14]. Global entity matching and span-based single-stage decoding provide further ways to reduce pipeline error propagation [15, 16]. These studies primarily improve the extraction architecture; our focus is the provenance and acceptance policy applied to generated candidates.

GLiNER and GLiREL provide label-conditioned alternatives for entity recognition and relation classification [17, 18]. These models differ from locally fine-tuned generators in supervision, pretraining, and output construction. We therefore report the implemented train-calibrated pipeline as a named external baseline, not as an exhaustive assessment of either model family. A same-backbone zero-shot generator further separates the benefit of local adaptation from that of graph conditioning.

2.2. Graph context and constrained generation

Knowledge graphs represent typed entities and relations explicitly, while retrieval augmentation uses selected records as model context [4, 5]. GenIE constrains generative information extraction with a structured output space and entity/relation catalogues [19]. Unified Structure Generation similarly maps heterogeneous information-extraction schemas to a common structured generation formulation [20]. Catalogue validity and source support are different requirements: an admissible entity or relation may still not be asserted in the current input. Our graph context is consequently treated as a candidate prior whose eligibility depends on training provenance and local source matching.

BGE-M3 supplies frozen text representations for semantic pattern retrieval [21]. It is neither a graph neural network nor a jointly optimized component of the extractor. QLoRA enables compact adaptation of the Qwen3 backbone [22, 23]. Neither method alone defines whether a candidate should enter the final assertion graph; the proposed gates make that acceptance policy explicit.

These distinctions separate three decisions: whether a graph record is eligible as context, whether a model proposes a candidate, and whether that candidate satisfies the output rules. A vocabulary constraint governs the form of a prediction; a provenance constraint governs the origin of a hint. Our framework combines the latter with a post-generation acceptance mechanism that checks source linkage and endpoint consistency. This allows candidate generation and deterministic selection to be evaluated separately.

2.3. Evidence and safety-oriented knowledge construction

Hallucination research distinguishes well-formed outputs from source-faithful ones [3]. In safety-related graph construction, prompt-based processing of accident reports likewise illustrates the need to trace generated knowledge back to its sources [24]. ADE offers a more controlled benchmark for a drug-safety-related relation task [6]. The present study uses its released relation annotations to evaluate source-linked extraction and the implemented evidence constraints.

The present study therefore treats extraction accuracy and inspectability as distinct evaluation targets. Benchmark annotations assess the predicted entities and relations, while source spans and gate decisions expose the acceptance path. Their combination is the focus of the proposed method; sentence-level evidence binding is not treated as a semantic truth test or as a substitute for human review.

3. Method

3.1. Task formulation

Let the annotated corpus be

$$\mathcal{D} = \{(x_i, \mathcal{E}_i, \mathcal{R}_i)\}_{i=1}^{N}, \tag{1}$$

where x_i is the deterministic single-space concatenation of the released tokens, $\mathcal{E}_i$ is its entity set, and $\mathcal{R}_i$ is its directed relation set. An entity mention is represented as

$$e = (b, u, y), \qquad x[b : u] = m, \tag{2}$$

where $[b, u)$ is a character interval, y is an entity label, and m is the copied mention text. A relation $r = (e_s, z, e_t)$ has typed directed endpoints and relation label z. Source identifiers and evidence spans are stored with these objects. Repeated strings at different offsets remain distinct mentions. The character-span requirement in Equation (2) prevents a normalized name alone from receiving credit for the wrong occurrence.

For a bounded graph context C_i, the extraction task is

$$f_\theta(x_i, C_i) = (\hat{\mathcal{E}}_i, \hat{\mathcal{R}}_i), \tag{3}$$

where both predicted sets must be reconstructed against x_i before they can be scored or accepted. Setting $C_i = \emptyset$ recovers the source-only input condition; nonempty context remains advisory and does not bypass the evidence gates.

The imported benchmark assigns every relation the status `explicit` and uses its source sentence as the evidence span. This representation keeps each evaluated relation tied to a concrete source sentence and makes the evidence path available for inspection.

3.2. Framework overview

Figure 1 summarizes the implemented flow. SOE (Source-Only Extraction), EAE (Exact-Anchor Extraction), and HRGE (High-Recall Graph Extraction) are separately trained QLoRA adapters on the same Qwen3-4B snapshot. SOE uses source text and the training-derived ontology without graph hints. EAE receives conservative concept hints, while HRGE receives bounded anchors, edge priors, and semantic relation patterns.

EVGE (Evidence-Verified Graph Extraction) applies the relation verifier to HRGE. CFE (Conservative-Fusion Extraction) pairs accepted HRGE entities with endpoint-filtered raw HRGE relations. PGE (Provenance-Gated Extraction) pairs the same accepted entities with endpoint-filtered verified relations. These three systems are deterministic post-processing variants, not additional trained networks. In deployment PGE needs EAE and HRGE candidates; SOE is a comparison branch and does not participate in fusion.

3.3. Provenance-controlled graph context

Each training graph stores mention strings, types, directed relations, and the sentence identifiers supporting them. For a training sentence d, a candidate is excluded if its only supporting provenance is d. Candidates with other supporting sentences remain eligible; their stored aggregate support statistics are not recomputed as a fully leave-one-out graph. This distinction prevents interpreting the provenance policy as complete removal of every contribution from the current example.

Formally, let $\mathcal{G}_{\text{tr}} = (\mathcal{V}_{\text{tr}}, \mathcal{A}_{\text{tr}})$ be the graph constructed from training annotations and let $\pi(c)$ denote the set of training-sentence identifiers supporting a context item c. During adapter training, the provenance-eligible pool for current training item i is

$$C_i^{\text{elig}} = \{c \in C_i^{\text{raw}} : \pi(c) \setminus \{i\} \neq \emptyset\}. \tag{4}$$

Evaluation identifiers are disjoint from training identifiers. Exact source-text overlap is handled separately by the quarantine rule below, which clears the entire context rather than treating duplicated text as independent evidence.

EAE uses type-balanced, exact-matching concept hints. HRGE uses three context channels. First, source-matched anchors must pass type-purity and mention-form filters. Second, a typed edge prior requires support outside the current example and both endpoint strings in the current source. Third, frozen BGE-M3 retrieves semantically related relation-pattern records. If $\mathbf{H}$ contains normalized eligible record embeddings and $\mathbf{q}$ is the normalized source embedding, retrieval scores are

$$\mathbf{s} = \mathbf{H}\mathbf{q}, \qquad \mathcal{I} = \text{TopK}_k\{i : s_i \geq \tau\}. \tag{5}$$

Equation (5) ranks candidate context, not accepted facts. The bounded retrieval limits and thresholds are fixed in the protocol artifact rather than tuned on the test set. The prompt explicitly identifies retrieved objects as hints. Exact-overlap quarantine overrides retrieval and clears all graph context for the affected evaluation sentences.

3.4. Structured generation and span reconstruction

The generator emits entity identifiers, copied text and types, plus relation identifiers, directed endpoint references, types, and explicit status. It is not asked to calculate source offsets. Deterministic expansion assigns copied mentions to occurrences in the supplied sentence and reconstructs character spans; unmatched strings and invalid references are recorded. This separates language-model candidate generation from source-coordinate bookkeeping.

Inference uses the full supplied sentence and a fixed generation policy. Failed jobs are represented by empty predictions, not removed from the denominator. These experiments do not use annotation-guided rescue windows or label-dependent sentence truncation.

3.5. QLoRA adaptation and training objective

All three generators use the same frozen Qwen3-4B backbone and a matched QLoRA adaptation configuration. Low-rank updates are applied to the attention and feed-forward projections. For an adapted projection ℓ, QLoRA parameterizes the effective weight as

$$W_\ell = W_\ell^{(0)} + \frac{\alpha}{r} B_\ell A_\ell, \qquad \text{rank}(B_\ell A_\ell) \leq r, \tag{6}$$

where $W_\ell^{(0)}$ is the frozen quantized weight, r is the adapter rank, and A_ℓ and B_ℓ are the trainable low-rank factors.

Let $y_{1:T}$ denote the serialized target and let ω_t equal one for a target token and zero for a prompt token. The optimized objective is

$$\mathcal{L}(\theta) = -\frac{1}{\sum_{t=1}^{T} \omega_t} \sum_{t=1}^{T} \omega_t \log p_\theta(y_t \mid y_{<t}, x, C). \tag{7}$$

Thus prompt tokens are excluded from the loss.

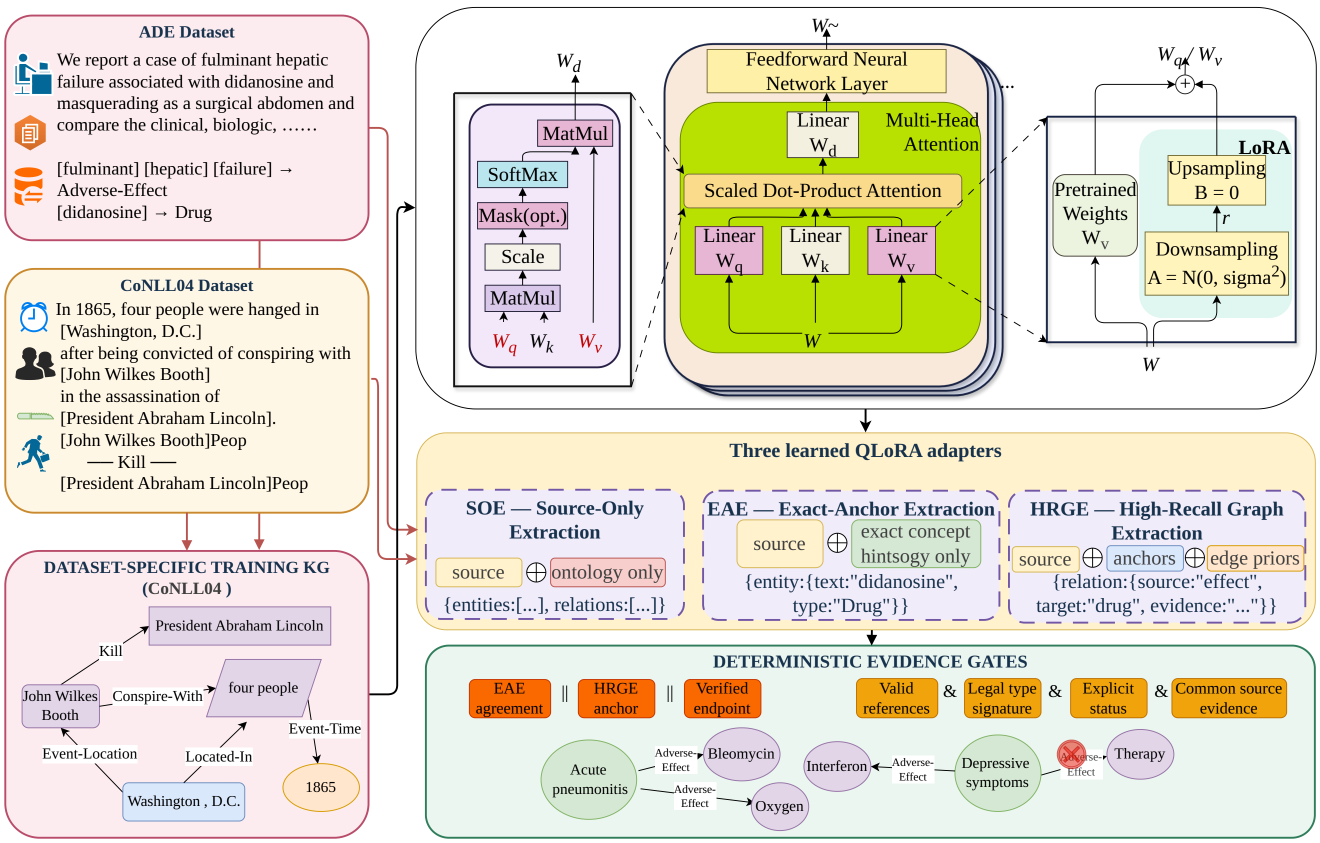

Figure 1: Detailed methodology used in this study. The figure shows the ADE and CoNLL04 inputs, dataset-specific training knowledge graphs, the Transformer attention block, the SOE/EAE/HRGE QLoRA adapter branches, and deterministic evidence gates based on agreement, anchors, endpoints, relation validity, and source evidence.

3.6. Provenance-aware candidate validation and fusion

For each relation label z, training annotations define permitted source-type and target-type sets S_z and T_z. The implemented type constraint is $y_s \in S_z$ and $y_t \in T_z$; it does not additionally require that every specific source/target type pair has been observed together. With component checks for valid references, label, endpoint types, explicit status, and source evidence, the relation predicate is

$$V(r) = V_{\text{ref}}(r) \wedge V_{\text{label}}(r) \wedge V_{\text{type}}(r) \wedge V_{\text{status}}(r) \wedge V_{\text{evidence}}(r). \tag{8}$$

Here V_{evidence} requires valid evidence and both endpoint strings in a common source-evidence span. The accepted relation set is

$$R_V = \{r \in R_H : V(r) = 1\}. \tag{9}$$

The entity gate uses three Boolean signals:

$$A(e) = A_{\text{agree}}(e) \vee A_{\text{anchor}}(e) \vee A_{\text{endpoint}}(e). \tag{10}$$

The corresponding accepted entity set is

$$E_A = \{e \in E_H : A(e) = 1\}. \tag{11}$$

Agreement means that EAE predicts the same case-folded, whitespace-normalized text and type. Anchor support requires a same-type source-gated HRGE anchor. Endpoint support means that the entity participates in a relation retained by Equation (9). The three signals in Equation (10) are complementary but not statistically independent; verified endpoints are a rule-based signal, not a second human judgement. Duplicate normalized text/type candidates are collapsed within the sentence. Relations are retained only when both referenced entities survive:

$$F(R, E) = \{(e_s, z, e_t) \in R : e_s \in E \wedge e_t \in E\}. \tag{12}$$

The two output compositions are

$$G_{\text{CFE}} = (E_A, F(R_H, E_A)), \qquad G_{\text{PGE}} = (E_A, F(R_V, E_A)). \tag{13}$$

Equation (13) makes the difference between raw-relation fusion and verified fusion explicit. Accept/reject decisions are logged. A retained relation satisfies the stated structural and source-linkage rules, and the applied checks are recorded with the output for inspection.

4. Experiments

The experiments connect the three contributions to observable outcomes: split-overlap controls examine

Table 1
Dataset inventory for the exact experimental partitions. Length is the number of whitespace-delimited tokens in the reconstructed source text.

Dataset	Split	Sentences	Entities	Relations	Mean length
ADE	train	3461	8717	5470	21.17
ADE	validation	384	996	627	21.82
ADE	test	427	1126	724	21.24
CoNLL04	train	922	3377	1283	28.77
CoNLL04	validation	231	893	343	30.27
CoNLL04	test	288	1079	422	28.94

provenance eligibility, compliance checks examine acceptance rules, and matched comparisons and ablations examine relation-extraction gains and component effects.

4.1. Experimental setup

ADE evaluates the primary medical application; CoNLL04 tests the same method in a supplementary general-domain setting. Each dataset has its own training graph and adapted generators. This design examines applicability across two domains under local adaptation, not direct transfer of a medically trained model to general-domain text. The experiments compare PGE with its matched source-only generator under a common strict-span protocol and examine component effects, training behavior, run-to-run stability, and exact-overlap sensitivity. The two-benchmark scope is a focused retrospective analysis rather than a preregistered representative sample of extraction domains. Model and gate settings were fixed before the reported test evaluation. Two additional training repetitions assess whether the observed gains depend on a single training run.

4.1.1. Datasets

We use the sentence-level ADE and CoNLL04 files distributed in SpERT format. Counts refer to that preprocessed distribution rather than every sentence in the original source corpora. CoNLL04 retains its supplied training, development, and test partitions. The ADE files contain 3,845 training sentences and 427 test sentences; a deterministic development slice of 384 sentences is removed from the supplied training pool, leaving 3,461 training sentences. The slice is selected by a fixed identifier-based procedure, not by validation performance. This is one fixed ADE split, not an average over the multiple folds sometimes used in the literature.

Table 1 gives counts recomputed from the exact imported artifacts. The unit called a document in the software is one benchmark sentence, not a whole medical case report or news article. Both datasets are English; the results therefore support neither multilingual nor cross-lingual claims. Published annotations are imported rather than locally re-annotated.

ADE contains DRUG and ADVERSE EFFECT entities and a single ADVERSE EFFECT relation label. The imported endpoint convention is *adverse effect to drug*; we preserve it rather than reversing it to a more intuitive drug-to-effect direction. CoNLL04 contains LOCATION, ORGANIZATION, PERSON, and OTHER entities, with WORK FOR, KILL, ORGANIZATION BASED IN, LIVE IN, and LOCATED IN relations.

Table 2 lists exact label counts for every partition. The single ADE relation label eliminates inter-class relation confusion but does not eliminate endpoint detection, direction, or false-pair errors. CoNLL04 has a broader relation inventory and more heterogeneous entity roles. These challenges motivate the proposed source-span reconstruction, type checks, and endpoint-consistent acceptance mechanism: identifying plausible entities alone does not determine which directed pairs should enter the output.

The mean training lengths are 21.17 reconstructed tokens for ADE and 28.77 for CoNLL04. Their test means are 21.24 and 28.94, respectively. The cumulative distributions describe all three partitions rather than only the training set. These lengths are descriptive whitespace-token counts, not Qwen tokenizer counts. The largest observed reconstructed length is 97 for ADE and 118 for CoNLL04 across the imported partitions.

Every imported sentence contains at least one annotated relation. The selected benchmark files therefore do not measure false-positive behavior over a deployment stream dominated by sentences with no relevant relations. We retain the supplied benchmark composition and do not create additional negative sentences for the reported tests.

Figure 2 reports training-label proportions and empirical cumulative sentence-length distributions for every partition. CoNLL04 is consistently shifted toward longer sentences, whereas the train, validation, and test curves within each dataset remain broadly aligned.

The graph is built separately for each dataset from its training annotations only. An exact-overlap check compares case-folded source strings after collapsing whitespace. ADE has no train/development or train/test exact-text overlap under this check. CoNLL04 has 14 development sentences and 19 test sentences whose text matches training content. For these sentences, both exact-anchor and high-recall graph context are cleared before generation.

Clearing graph hints does not undo exposure of the trained generator to duplicate training text. We therefore retain the supplied partitions for the main comparison and

Table 2
Exact entity and relation annotation counts in the imported files.

Dataset	Kind	Label	Train	Validation	Test
ADE	Entity	ADVERSE EFFECT	4631	529	616
ADE	Entity	DRUG	4086	467	510
ADE	Relation	ADVERSE EFFECT	5470	627	724
CoNLL04	Entity	LOCATION	1219	322	427
CoNLL04	Entity	ORGANIZATION	616	170	198
CoNLL04	Entity	OTHER	455	118	133
CoNLL04	Entity	PERSON	1087	283	321
CoNLL04	Relation	KILL	179	42	47
CoNLL04	Relation	LIVE IN	330	91	100
CoNLL04	Relation	LOCATED IN	247	65	94
CoNLL04	Relation	ORGANIZATION BASED IN	271	76	105
CoNLL04	Relation	WORK FOR	256	69	76

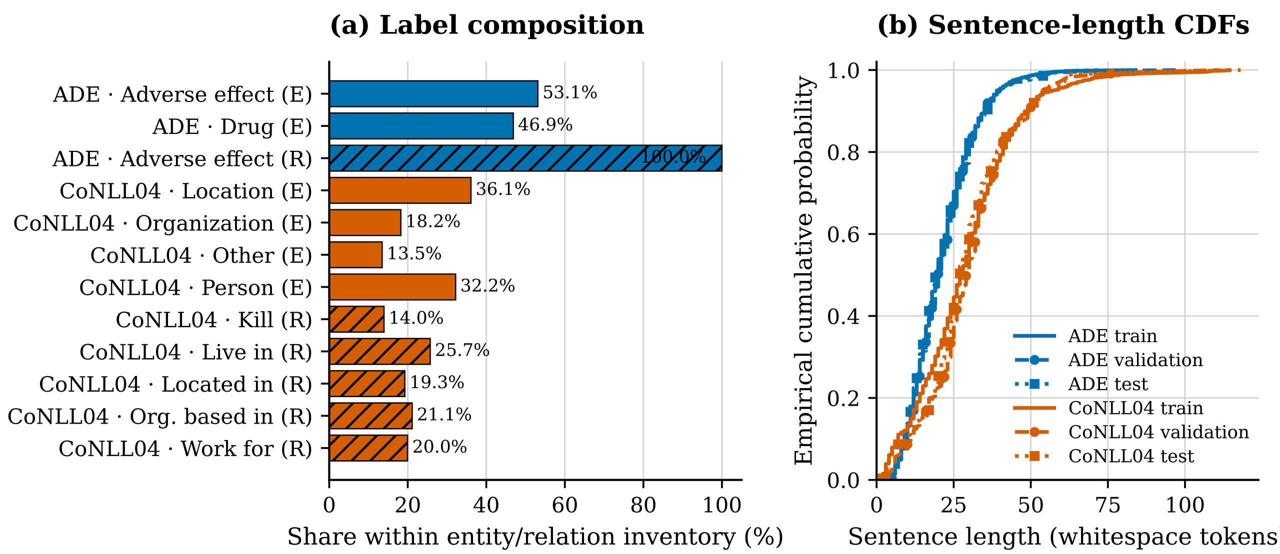

Figure 2: Dataset distribution analysis. (a) Training-label composition within each dataset's entity (E) or relation (R) inventory. Percentages sum to 100% separately within each dataset–kind combination. (b) Empirical cumulative distributions of reconstructed sentence length for the train, validation, and test partitions. Length is measured in whitespace-delimited tokens.

report an additional fixed-prediction test analysis that excludes the 19 overlapping CoNLL04 test sentences. We do not claim that the benchmark is duplicate-cluster isolated or that this exact-match check detects all near duplicates.

4.1.2. Baselines and metrics

The internal development comparison includes all six named systems. The test table includes the two promoted systems and three completed external comparisons. Qwen3-4B zero-shot uses the same backbone without local adapters, followed by the recorded source expansion and relation verifier. SpERT is trained from a base encoder on the same training partitions and its final checkpoint is used without selecting a test-best epoch. GLiNER+GLiREL uses predicted, not gold, entities and training-only calibration. These are heterogeneous baseline configurations, not equal-compute experiments.

All reported main-table predictions are rescored with one public character-span evaluator. An entity key contains its exact source interval and type. A relation key contains both typed directed entity keys and the relation label. Predicted evidence offsets must resolve to the supplied source; the evaluator does not recover missing offsets by searching normalized text. Duplicate keys receive one vote, invalid objects remain unmatched predictions, and omitted jobs remain empty. For pooled counts,

$$
\begin{aligned}
P &= \frac{TP}{TP+FP},\\
R &= \frac{TP}{TP+FN}, \qquad (14)\\
F_1 &= \frac{2TP}{2TP+FP+FN}.
\end{aligned}
$$

Equation (14) is applied across sentences within each dataset, not after averaging their individual F1 scores.

Because all imported statuses are explicit, claim-status-aware scores do not form a separate uncertainty benchmark.

4.1.3. Statistical analysis and compliance evaluation

We compute paired bootstrap resamples of test sentences from fixed predictions and recompute pooled F1 within each resample. The percentile interval describes sentence-sampling uncertainty for the completed model pair, not variability across training repetitions. Intervals are descriptive, unadjusted for multiple comparisons, and are not used to change the frozen models or gates.

To reduce reliance on a single training outcome, we conduct two additional training repetitions using the same data partitions and matched optimization settings. We report the mean and sample standard deviation of strict-span test F1 to assess whether the relation gains recur. These repetitions measure observed run-to-run stability rather than eliminating statistical dependence.

Evidence coverage and rule-invalid counts are measured separately from extraction F1. These compliance counts refer to emitted prediction objects before the evaluator's key deduplication. Exact-overlap sensitivity removes the same affected test sentence identifiers from both systems, without retraining or regenerating predictions. Test labels are used for reporting statistics and scoring only, not for constructing graph hints or choosing thresholds in this revision.

The compliance assessment distinguishes type and reference validity from source-evidence binding. Because the benchmark assigns the source sentence as evidence, endpoint co-occurrence establishes a traceable link but does not prove the predicted relation's meaning. We therefore evaluate rule compliance and gold-referenced extraction accuracy separately. The overlap analysis likewise tests whether the observed gain persists after known repeated texts are excluded, rather than assuming that context quarantine alone removes training exposure.

4.2. Main results

Table 3 reports the completed test results under the common evaluator. PGE raises ADE relation precision from 80.62% to 87.00% and recall from 67.82% to 72.10%, giving relation F1 of 78.85% compared with 73.67% for SOE. Entity F1 remains essentially unchanged, at 87.93% versus 87.97%. Thus, the main improvement is relational rather than a general gain in mention recognition.

On CoNLL04, PGE raises relation precision from 54.61% to 62.40% and recall from 56.16% to 57.82%. Relation F1 increases from 55.37% to 60.02%, while entity F1 decreases from 83.83% to 83.20%. The composition trades some entity coverage for a better relation precision–recall balance in this run.

Figure 3 complements the exact values in Table 3. It highlights that the matched PGE–SOE difference is relation-specific: relation F1 increases on both datasets, whereas the entity differences are close to zero and their paired intervals cross zero.

SpERT has the highest entity and relation F1 among these completed systems: 91.37%/84.08% on ADE and 89.02%/69.72% on CoNLL04. PGE therefore improves the matched generative baseline but does not surpass this trained span-based model. PGE also exceeds the recorded zero-shot and GLiNER+GLiREL pipelines on both extraction tasks. The latter observation is specific to their implemented label mapping and calibration; it is not a claim that every configuration of those model families would perform worse.

The acceptance mechanism produces zero rule-invalid relation objects in PGE on both tests, compared with 31 for SOE on ADE and 16 on CoNLL04. HRGE already has zero and three such objects, respectively, so these counts reflect both candidate quality and subsequent filtering. All three systems have zero measured unsupported-evidence objects under the sentence-level evidence check; the observed distinction is rule compliance, not a demonstrated reduction in semantic hallucination.

Excluding the 19 exact CoNLL04 train–test overlaps leaves 269 test sentences. PGE retains a relation F1 of 59.10%, compared with 53.79% for SOE, a 5.31-point gain. Thus the relation advantage persists beyond the known exact overlaps. ADE has no such overlaps under the same check.

4.3. Training-loss diagnostics

Figure 4 shows the EAE and HRGE training-loss curves. CoNLL04 exhibits a pronounced early reduction followed by smaller changes, whereas ADE starts at a lower loss and shows a modest late reduction amid fluctuations. The trajectories describe optimization of structured generation; the held-out comparisons separately assess extraction performance.

4.4. Additional-run stability

Table 4 summarizes the two later repetitions. On ADE, PGE improves relation F1 over SOE in both repetitions (81.20% versus 78.31%, and 80.62% versus 78.06%). On CoNLL04, the corresponding differences are positive but smaller (58.50% versus 56.91%, and 61.45% versus 60.92%). Across the two runs, the mean relation-F1 difference is 2.73 points on ADE and 1.06 points on CoNLL04. Relation-F1 gains over SOE therefore recur on both datasets in the two additional training runs.

4.5. Ablation study

Table 5 separates graph-conditioned candidate generation, relation verification, and entity-gated fusion on the development sets. HRGE–SOE comparisons assess the graph-conditioned generator as a whole; EVGE–HRGE comparisons isolate relation verification; PGE–EVGE comparisons assess entity selection with endpoint filtering. These comparisons explain component behavior without selecting variants on test results.

On ADE, graph conditioning raises relation recall from 68.26% for SOE to 72.89% for HRGE and relation F1 from 73.16% to 76.29%. Entity recall also rises from 82.53% to 85.54%. This joint increase is consistent with graph hints helping recover relevant mentions and pairs. The exact-anchor EAE branch instead reaches higher relation precision (80.84%) with lower recall (67.30%), illustrating the

Table 3
Completed test comparisons, strict source-character-span micro scores (%). Qwen3-ZS is Qwen3-4B zero-shot with the recorded verifier; GLiNER+GLiREL uses training-only calibration. Each dataset uses its own test partition. All rows are rescored with the same evaluator. Bold values marked with ↑ are the highest in each metric within each dataset.

Dataset	System	Ent. P	Ent. R	Ent. F1	Rel. P	Rel. R	Rel. F1
ADE	Qwen3-ZS + verifier	63.83	63.32	63.58	39.36	22.24	28.42
ADE	GLiNER + GLiREL	79.30	64.65	71.23	60.31	43.23	50.36
ADE	SpERT	90.65	**92.10 ↑**	**91.37 ↑**	82.19	**86.05 ↑**	**84.08 ↑**
ADE	SOE	92.21	84.10	87.97	80.62	67.82	73.67
ADE	PGE	**92.89 ↑**	83.48	87.93	**87.00 ↑**	72.10	78.85
CoNLL04	Qwen3-ZS + verifier	48.78	53.94	51.23	30.92	19.19	23.68
CoNLL04	GLiNER + GLiREL	60.78	14.37	23.24	13.79	3.79	5.95
CoNLL04	SpERT	**88.61 ↑**	**89.43 ↑**	**89.02 ↑**	**71.01 ↑**	**68.48 ↑**	**69.72 ↑**
CoNLL04	SOE	85.83	81.93	83.83	54.61	56.16	55.37
CoNLL04	PGE	86.04	80.54	83.20	62.40	57.82	60.02

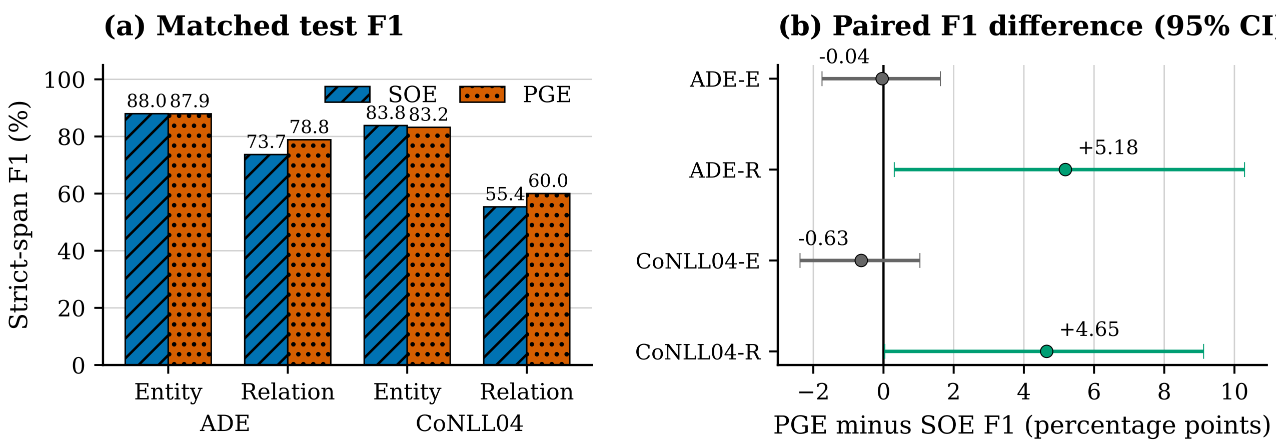

Figure 3: Test-set analysis of the matched SOE and PGE systems. (a) Strict-span entity and relation F1 on ADE and CoNLL04. (b) PGE-minus-SOE F1 differences; horizontal intervals are paired 95% sentence-bootstrap intervals from fixed predictions. Exact values and all external baselines remain in Table 3.

different candidate profiles available to fusion. EVGE and HRGE have identical scores: the verifier imposes no additional filtering on these already rule-valid candidates, so this ADE gain originates in candidate generation rather than verification.

ADE entity-gated fusion raises relation precision from 80.04% for EVGE to 81.39% for PGE while reducing recall from 72.89% to 71.13%. Relation F1 is consequently 75.91%, still 2.75 points above SOE but below HRGE. Since PGE retains a subset of the verified candidates, the

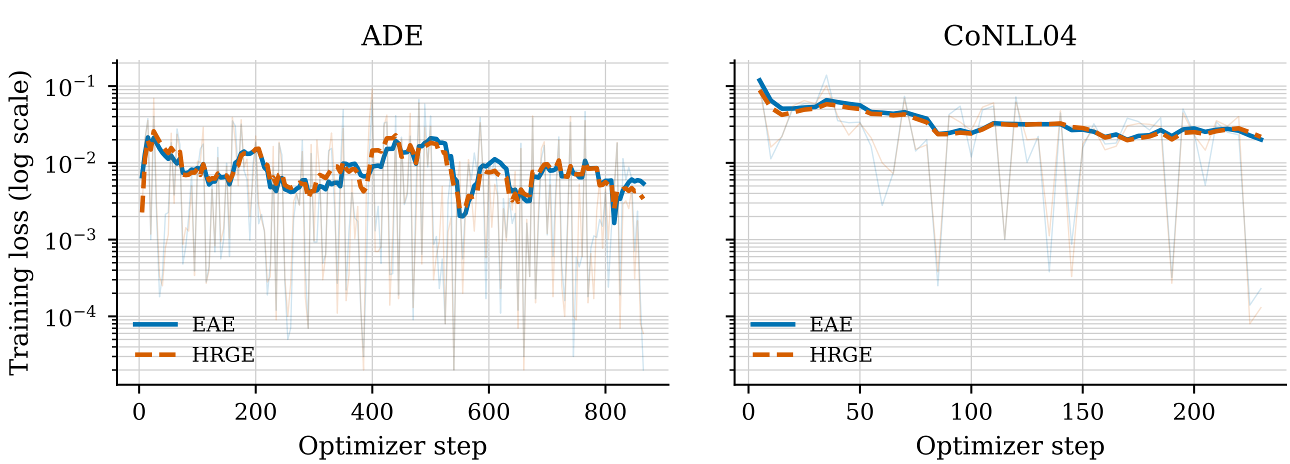

Figure 4: Training loss for the EAE and HRGE adapters. Faint lines show raw observations sampled every five optimizer steps; emphasized curves are ten-observation trailing means. The logarithmic scale preserves the variation in the raw observations.

Table 4
Test stability across two additional training repetitions under the common strict-span evaluator (micro %, mean ± sample standard deviation; $n = 2$). The final column is the mean PGE-minus-SOE relation-F1 difference.

Dataset	System	Entity F1	Relation F1	Δ relation F1
ADE	SOE	88.00 ± 0.37	78.18 ± 0.18	–
ADE	PGE	89.35 ± 0.04	80.91 ± 0.41	+2.73
CoNLL04	SOE	82.26 ± 1.91	58.92 ± 2.84	–
CoNLL04	PGE	81.75 ± 2.02	59.98 ± 2.09	+1.06

Table 5
Development-set ablation under the common strict-span evaluator (micro %). These rows characterize the fixed architecture and are not additional promoted test systems.

Dataset	System	Ent. P	Ent. R	Ent. F1	Rel. P	Rel. R	Rel. F1
ADE	SOE	89.25	82.53	85.76	78.82	68.26	73.16
ADE	EAE	90.49	81.22	85.61	80.84	67.30	73.46
ADE	HRGE	90.35	85.54	87.88	80.04	72.89	76.29
ADE	EVGE	90.35	85.54	87.88	80.04	72.89	76.29
ADE	CFE	90.48	83.94	87.08	81.39	71.13	75.91
ADE	PGE	90.48	83.94	87.08	81.39	71.13	75.91
CoNLL04	SOE	84.94	82.08	83.49	61.33	59.18	60.24
CoNLL04	EAE	83.71	82.87	83.29	58.10	60.64	59.34
CoNLL04	HRGE	82.35	81.52	81.94	60.06	56.56	58.26
CoNLL04	EVGE	82.35	81.52	81.94	62.99	56.56	59.60
CoNLL04	CFE	84.97	78.50	81.61	63.37	55.98	59.44
CoNLL04	PGE	84.97	78.50	81.61	65.08	55.98	60.19

recall loss indicates that some correct relations lose eligible endpoints. The precision gain and coverage loss identify a concrete limitation of the current entity selection rule: it filters incorrect candidates but can also reject valid ones.

On CoNLL04, HRGE relation F1 is 58.26%, compared with 60.24% for SOE. The broader relation inventory and more heterogeneous entity roles may make graph hints more ambiguous, although these factors are not isolated here. Relation verification raises precision from 60.06% to 62.99% without changing recall (56.56%), increasing F1 to 59.60%. Under this filtering-only comparison, the improvement is consistent with removing incorrect relations while retaining the correctly extracted ones.

The final CoNLL04 fusion raises relation precision further to 65.08%, with 55.98% recall and 60.19% F1. Compared with CFE, PGE improves relation F1 from 59.44% at the same recall, showing the benefit of verified rather than raw relations within the same accepted entity set. Entity precision rises from 82.35% for HRGE to 84.97% for PGE, but entity recall falls from 81.52% to 78.50%. The final relation F1 thus recovers most of HRGE's deficit and is close to SOE on this development set. Together, the ablations support complementary roles for candidate expansion and selective acceptance, while identifying endpoint retention as a target for improving coverage.

5. Discussion

5.1. Benchmark findings

The third contribution is supported by a relation-F1 improvement over source-only generation on both tests and in the two additional training runs. The gains arise alongside higher relation precision on both tests, while entity F1 stays similar or declines slightly. This is consistent with an architecture designed to govern candidate acceptance rather than simply maximize output volume. The validation ablation further shows that graph conditioning and selection interact: recall-oriented context can help on ADE but does not automatically improve CoNLL04, and a verifier may be inactive when candidates already satisfy its rules.

The distribution analysis helps interpret these differences. ADE has a larger training pool, shorter average sentences, and a single relation class. CoNLL04 has fewer training sentences, longer inputs, and five relation labels. These factors are confounded rather than experimentally isolated. Their association with the observed scores motivates separate dataset reporting; the factors should not be conflated in a single pooled interpretation.

The focused, retrospectively narrowed scope and small number of completed training repetitions limit generalization. The fixed ADE partition is not directly interchangeable with published multi-fold results. Both corpora are English and contain relation-bearing sentences, so they do not establish multilingual behavior, long-report performance, or false-positive rates on unrestricted streams. Sentence-level bootstrap intervals also cannot recover dependence between sentences from the same upstream article when article identifiers are unavailable.

5.2. Auditability

The first two contributions govern different points in the extraction path: provenance controls determine which graph hints are eligible, whereas the acceptance gates determine which generated candidates enter the output. Graph support identifies why a candidate was suggested; reconstructed offsets identify where it appears; gate records identify why it was retained or rejected. However, this benefit is architectural. No user study in the present experiments measures reduced analyst effort, improved decisions, or greater trust. It would be inappropriate to substitute perfect rule compliance for such evidence.

The gates also impose a coverage cost. Agreement operates on normalized text/type pairs, and repeated mentions may be collapsed during fusion even though the evaluator distinguishes offsets. This can remove legitimate occurrences. A legal endpoint signature and sentence-level co-occurrence are insufficient to prove a relation's meaning, and three gate signals should not be interpreted as three independent confirmations. These observations explain why the final output remains a candidate assertion graph for review.

Exact-overlap quarantine prevents direct graph hints for known repeated source text but does not remove the generator's training exposure. The sensitivity analysis excludes known exact duplicates only, and pretrained components may have encountered related material before task adaptation. The available artifacts cannot rule out such exposure. Moreover, the ontology's separate source- and target-type sets are less restrictive than a catalogue of observed joint type pairs.

5.3. Safety-related implications

The primary medical application is drug-safety information extraction on ADE; CoNLL04 supplements this evaluation with general-domain relational structure. The framework can present an analyst with the candidate relation, source sentence, typed endpoints, and provenance of its acceptance path. An analyst can then accept, amend, or reject the proposed association. Absence from the retained graph is not evidence of the absence of an adverse effect or another real-world relation.

The reported task focuses on source-linked extraction and evidence-path inspection. The framework is intended to support analyst review by making the origin and acceptance path of each extracted assertion explicit.

Generative adaptation, supervised SpERT training, and pretrained label-conditioned models use different training regimes. PGE requires two adapted generators plus frozen retrieval, so compact adaptation must not be confused with measured end-to-end cost savings. No new clinical annotations or independent evidence-judgement labels were collected for this paper. Public availability of a dataset or model does not by itself permit every downstream redistribution or commercial use; original licenses remain applicable.

6. Conclusion and future work

This study introduces provenance-constrained graph augmentation and an evidence-gated acceptance mechanism for auditable drug–adverse-effect extraction from medical text, with a supplementary general-domain evaluation. The mechanism decouples candidate generation from output acceptance. Training-source eligibility controls the use of graph hints, while source-span reconstruction and entity–relation checks expose the acceptance path of retained outputs. Under a unified strict-span evaluator, the completed PGE tests improve relation F1 over SOE by 5.18 percentage points on ADE and 4.65 points on CoNLL04, while offering an explicit source-linked acceptance path. The CoNLL04 relation advantage remains after exact train/test overlaps are excluded, and two additional runs retain positive relation-F1 differences. Ablations distinguish the contribution of graph-conditioned candidate recovery from that of selective acceptance and identify endpoint retention as a precision–coverage trade-off. The outcome is a controlled, auditable graph-informed extraction layer that keeps generated entities and relations strongly connected to their source text and exposes the acceptance path for human review.

Future work should test the framework on preregistered multi-fold splits, multilingual and long-document corpora, and deployment-like streams containing many sentences with no relevant relation. Near-duplicate grouping can extend the present exact-overlap analysis, while calibrated gate confidence and joint source–target type constraints may improve the precision–coverage trade-off. Larger repeated-run studies, end-to-end latency and resource measurements, and analyst studies are also needed to determine whether inspectable acceptance paths reduce review effort or improve downstream decisions.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence this work.

Data and code availability

The data and code supporting the findings of this study are available from the corresponding author upon reasonable request.

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